Mediants and Collatz: Constructing Generalised Collatz Trajectories via Rational Approximation

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Abstract

We describe an application of the mediant tree algorithm of [1] to the construction of generalised Collatz trajectories. Given a generalised Collatz system (g, h) with odd step $x \mapsto gx + q$ and even step $x \mapsto x/h$, a parity sequence realising a good rational approximant to $\theta = \log_h g$ is produced directly by the mediant tree as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$. We show how to convert such a mediant word into a concrete starting integer x_0 via a standard modular-arithmetic pipeline, and explain why the canonical greedy parity-sequence algorithm produces exactly the Sturmian word for slope $1/(1+\log_h g)$. We compare our approach to Eliahou's continued-fraction bounding method [2], and note that the mediant tree simultaneously produces both the rational approximant and a concrete trajectory realising it — a capability absent from Eliahou's framework.

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1 Introduction

A *generalised Collatz sequence* is defined by a pair (g, h) with $g, h \geq 2$ and $\gcd(g, h) = 1$: an odd element x is mapped to $gx + q$ and an even element x is divided by h. The standard Collatz sequence corresponds to $g = 3$, $h = 2$, $q = 1$. The parity structure of a trajectory — the sequence of odd and even steps — is governed by $\theta = \log_h g$: a long-descending trajectory must have the ratio of even to odd steps approximate θ from above.

The companion paper [1] develops the *mediant tree*, an algorithm that constructs increasingly precise rational approximants to any real $\theta > 1$ together with a binary string over $\{\mathsf{O}, \mathsf{E}\}$ — the *mediant word* — that encodes the approximation path. The mediant word is proven to be identical to the Sturmian word with slope $1/(1 + \theta)$ at all interior positions, establishing a tight connection between rational approximation and combinatorics on words.

The present paper describes the Collatz application that originally motivated the development of [1]. The main contributions are:

1. A pipeline (Section 4) for converting a mediant word into a concrete starting integer x_0 whose generalised Collatz trajectory follows the prescribed parity pattern for exactly $e + o$ steps.

2. A description (Section 5) of the greedy parity-sequence algorithm and a proof that it produces the Sturmian word for slope $1/(1 + \log_h g)$.

3. A discussion (Section 7) of the relationship to Eliahou's work on rational approximation of $\log_2 3$ in the context of Collatz cycle bounds.

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Organisation. Section 2 recalls the key definitions and results from [1]. Section 3 introduces generalised Collatz sequences and parity sequences. Section 4 presents the step-by-step pipeline from mediant word to Collatz trajectory. Section 5 describes the greedy algorithm and its Sturmian interpretation. Section 6 discusses why general Collatz trajectories are not Sturmian. Section 7 compares with Eliahou's approach.

2 Preliminaries

We recall the definitions and results from [1] that are used in this paper. All notation is consistent with that reference.

A *ratio* is a pair $(e, o) \in \mathbb{N}^2$ with $\gcd(e, o) = 1$, written e/o. Given a real $\theta > 1$, a ratio $r = e/o$ is a *good estimate* of θ if $|e - \theta o| < 1$. Two good estimates $r_i = e_i/o_i$ and $r_{i+1} = e_{i+1}/o_{i+1}$ form a *bounded pair* (r_i, r_{i+1}) if they bracket θ and satisfy $\Delta(r_i, r_{i+1}) := e_{i+1}o_i - e_i o_{i+1} = \pm 1$.

Given a bounded pair (r_i, r_{i+1}) and a weight $w \geq 1$, the *weighted mediant* is

$$\mathrm{Mediant}(r_i, r_{i+1}, w) = \frac{e_i + w\, e_{i+1}}{o_i + w\, o_{i+1}}.$$

The *optimal weight* w^* is the unique integer such that the pair $\big(\mathrm{Mediant}(r_i, r_{i+1}, w^* - 1), \mathrm{Mediant}(r_i, r_{i+1}, w^*)\big)$ brackets θ; it is given in closed form by Theorem 8.1 of [1].

The *mediant word* $\pi(r)$ of a ratio r produced by the mediant tree is a string over $\{\mathsf{O}, \mathsf{E}\}$ of length $e + o$ encoding the sequence of weighted-mediant choices made to reach r from the initial bounded pair. It is a lower Christoffel word (Lemma 9.3 of [1]), begins with EE for every non-leaf estimate (Theorem 12.4 of [1]), and is identical to the Sturmian word for slope $1/(1+\theta)$ at all interior positions (Theorem 12.6 of [1]).

3 Generalised Collatz Sequences and Parity Sequences

A *generalised Collatz sequence* uses an odd step $x \mapsto gx + q$ and an even step $x \mapsto x/h$, where $g, h \in \mathbb{N}$ with $g, h \geq 2$ and $\gcd(g, h) = 1$, and $q \in \mathbb{Z}$ with $q \neq 0$. The parity sequence structure depends only on g and h through $\theta = \log_h g$; the value of q affects only the concrete starting residue x_0 (Step 4 of the pipeline in Section 4). A *parity sequence* is a string over $\{\mathsf{O}, \mathsf{E}\}$ recording the sequence of odd and even steps along a trajectory; O denotes an odd step and E an even step.

For a trajectory to descend on average, the ratio of even steps to odd steps must exceed $\log_h g$: if a segment of length $e + o$ has e even steps and o odd steps, the net effect is multiplication by g^o/h^e, which is less than 1 iff $e/o > \log_h g$. The boundary case $e/o = \lceil o \log_h g \rceil / o$ is a good estimate of $\log_h g$, and corresponds to the upper-bound condition

$$h^{e-1} < g^o < h^e.$$

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A parity sequence realising a good estimate e/o must therefore have exactly o copies of O and e copies of E, with no two consecutive O's. OO-avoidance holds on three independent grounds: it is a property of Sturmian sequences with $\alpha < 1/2$ (equivalently $\theta > 1$); it holds for $h = 2$ trajectories when q is odd and $\gcd(g, h) = 1$; and it is maintained by the mediant word generator by construction. By contrast, for $h \neq 2$ arbitrary trajectories need not be OO-free. Every OO-free parity sequence can be realised as a $gx + q$, x/h trajectory for some q determined by the sequence [3].

4 From Mediant Word to Collatz Trajectory: the Pipeline

The mediant tree applies to any real $\theta > 1$, not only to logarithms. The following pipeline converts a mediant word into a concrete Collatz starting integer for any (g, h) with $\theta = \log_h g$.

Step 1: Choose a real θ and an initial bounded pair. Select any real $\theta > 1$ and a Farey-adjacent bounded pair (r_0, r_1) with $r_0 < \theta < r_1$. For the standard Collatz system take $\theta = \log_2 3$; for the generalised system (g, h) take $\theta = \log_h g$. The construction applies equally to any irrational θ (e.g. π, $\sqrt{2}$), in which case it produces rational approximants without a Collatz interpretation.

Step 2: Run the mediant tree. Apply the optimal weight formula (Theorem 8.1 of [1]) to produce a sequence of mediant words $\pi(r_{2n+1})$ of increasing length. At level n, the word $\pi(r_{2n+1})$ has o_{2n+1} odd steps and e_{2n+1} even steps, and e_{2n+1}/o_{2n+1} is a good estimate of θ satisfying $h^{e-1} < g^o < h^e$ (upper bound) or $h^e < g^o < h^{e+1}$ (lower bound).

Step 3: Derive the path identity. For a chosen mediant word $\sigma = \pi(r_{2n+1})$, the function `path_identity(`σ`)` constructs the *basic path identity*

$$h^e\, x_n = g^o\, x_0 + q\, K(g,h),$$

where x_0 is the starting value, x_n is the value after $n = e + o$ steps following σ, and $K(g,h)$ is the *carry polynomial* — a polynomial in g and h whose coefficients record the cumulative effect of the additive terms along the trajectory, with q scaling the carry to the chosen additive constant. This identity holds for *every* x_0; it is a universal statement about the linear map defined by σ. Adding $g^o h^{e+1} t$ to both sides for any integer t gives the equivalent *shifted path identity*

$$h^e\big(x_n + g^o h\, t\big) = g^o\big(x_0 + h^{e+1} t\big) + q\, K(g,h),$$

which makes explicit that the family of valid (x_0, x_n) pairs forms an arithmetic progression with step $(h^{e+1}, g^o h)$, motivating the modulus h^{e+1} in Step 4.

Step 4: Instantiate to concrete (g,h) and solve for x_0. The function `path_identity_params(`σ`, `g`, `h`)` substitutes the concrete multiplier and base, computes K, and solves

$$x_0 \;\equiv\; -q\, K \cdot (g^o)^{-1} \pmod{h^{e+1}}$$

for the canonical starting residue $x_0 \in [0, h^{e+1})$ whose trajectory follows σ for exactly $o + e$ steps. The modulus is h^{e+1} rather than h^e because the word ends in O: after all e halvings the final odd step imposes one additional divisibility condition on x_0. For $h = 2$ (the standard Collatz system) this residue is unique; for $h \geq 3$ there may be multiple valid starting residues modulo h^{e+1}, and the formula gives the canonical (smallest non-negative) representative.

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Step 5: Read off the approximation data. The output tuple $(o,\ e,\ e/o,\ x_0)$ encodes:

- e/o — a rational approximant to θ, with $e/(o+e)$ and $o/(o+e)$ the even and odd step densities;
- x_0 — the canonical starting residue (mod h^{e+1}) whose $gx + q$ trajectory realises the mediant word σ for exactly $o + e$ steps.

Setting $t = 0$ gives $x_n = (g^o x_0 + qK)/h^e$, which is also determined by x_0. As $n \to \infty$ the ratio e_{2n+1}/o_{2n+1} converges to θ, with convergence rate governed by the continued-fraction expansion of θ.

Remark. Evaluating the Collatz iteration from x_0 is *not* an efficient way to generate the parity sequence: it requires arithmetic with integers of size $O(h^e)$, which grows exponentially in the word length. The parity sequence itself is produced directly and efficiently by the mediant tree, with no large integer arithmetic. The value of x_0 lies in establishing that a concrete Collatz trajectory realising the sequence exists — not in using it to regenerate the sequence.

Remark (residue constraint). To the extent that a $3x + 1$ trajectory matches a factor of the Sturmian word for $\theta = \log_2 3$, every odd element of the trajectory is congruent to 1 or 3 modulo 8. This is a consequence of the Sturmian alphabet being $\{\mathsf{EO}, \mathsf{EEO}\}$: an odd

step from a residue $r \equiv 1 \pmod 8$ produces $v_2(3r+1) = 2$ even steps (an EEO block), while $r \equiv 3 \pmod 8$ produces $v_2(3r+1) = 1$ even step (an EO block); residue 5 (mod 8) forces the parity sequence OEEE$\cdots$, and residue 7 (mod 8) forces the parity sequence OEOEO; neither pattern occurs in the Sturmian word for $\log_2 3$, so neither residue can appear in a trajectory matching it. A full account is given in [4].

5 The Greedy Parity Sequence Algorithm

For any generalised Collatz system (g, h) with $g, h \geq 2$ and $\gcd(g, h) = 1$, there is a simple greedy algorithm that generates the idealised parity sequence using only integer arithmetic — no floating-point evaluation of $\log_h g$ is required. Let $short = \lfloor \log_h g \rfloor$ and $long = short+1$ (the two possible E-run lengths, since $\log_h g \in (short, long)$).

```
def collatz_log_h_of_g(g=3, h=2):
    e=o=0
    short=math.floor(math.log(g, h))
    long=short+1
    while True:
        if g**(o+1) < h**(e+short):
            yield from 'E' * short
            yield 'O'
            e+=short
            o+=1
        else:
            yield from 'E' * long
            yield 'O'
            e+=long
            o+=1
```

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The condition $g^{o+1} < h^{e+short}$ is a pure integer comparison equivalent to $(o+1)\log_h g < e + short$, i.e. appending a *short* block $\mathsf{E}^{short}\mathsf{O}$ is sufficient to keep the ratio e/o above $\log_h g$; otherwise a *long* block $\mathsf{E}^{long}\mathsf{O}$ is needed. This is exactly the Beatty/Sturmian greedy rule: at each O step the algorithm selects the shorter run when the fractional part of $(o+1)\log_h g$ is already covered by the accumulated e, and the longer run otherwise.

The resulting sequence is the Sturmian word for slope $\alpha = 1/(1 + \log_h g)$, as verified by the identity $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ for all n.

For the standard case $g = 3$, $h = 2$ the first 60 symbols are:

EEOEEOEOEEOEOEEOEEOEOEEOEOEEOEEOEOEEOEOEEOEOEEOEEOEOEEOEOEEO,

with O-density converging to $1/(1 + \log_2 3) = \log_6 2 \approx 0.387$ — the *ideally balanced* parity pattern, descending as quickly as theory permits. For $g = 5$, $h = 2$ ($short = 2$, $long = 3$, since $\log_2 5 \approx 2.322$) the first 60 symbols are:

EEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEO,

with O-density converging to $1/(1+\log_2 5) \approx 0.301$. The $5x+1$ system has known non-trivial cycles; the mediant word describes the idealised parity pattern of a trajectory that descends at the asymptotically optimal rate for that system.

6 General Collatz Sequences are not Sturmian

The Sturmian word for slope $\alpha = 1/(1+\theta)$ consists of exactly two block types: $\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil\theta\rceil}\mathsf{O}$, interleaved in the Sturmian pattern. These are precisely the leaf words of the mediant tree for θ. A Sturmian word can never contain a block of any other length.

A general Collatz trajectory is far less constrained: an actual orbit can produce E-runs of any length, interleaved freely. The mediant word (equivalently, the Sturmian word) is the *ideally balanced* special case.

Moreover, even among trajectories that approximate $\log_h g$ closely, the Sturmian alphabet $\{\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}, \mathsf{E}^{\lceil\theta\rceil}\mathsf{O}\}$ is not the only primitive alphabet that can arise. For example, the known non-trivial cycles of the $5x+1$ system have alphabets $\{\mathsf{EO}, \mathsf{EEEEO}\}$, $\{\mathsf{EO}, \mathsf{EEEEEO}\}$, and $\{\mathsf{EO}, \mathsf{EEEO}\}$ — each containing the block EO of length 2, which does not appear in the Sturmian alphabet $\{\mathsf{EEO}, \mathsf{EEEO}\}$ at all. The mediant tree generates parity sequences over the Sturmian alphabet only; it produces the optimal approximation to θ but does not exhaust all primitive alphabets whose e/o ratio approximates θ.

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7 Relationship to Eliahou's Work

Eliahou [2] studies rational approximation of $\log_2 3$ in the context of bounding nontrivial cycle lengths in the Collatz problem. His approach identifies best upper rational approximants via the continued-fraction expansion of $\log_2 3$, yielding explicit lower bounds on Collatz cycle lengths.

Our work differs in two respects. First, our algorithm constructs approximants directly from bounded pairs via the optimal weight formula (Theorem 8.1 of [1]), with no continued-fraction computation; the connection to continued fractions is a consequence of the Sturmian identity. Second, and more fundamentally, Eliahou's framework gives no means of constructing a parity sequence realising a given approximant. The mediant tree simultaneously produces the rational approximant and the mediant word that realises it, as a byproduct of the same construction.

Where Eliahou's work illuminates the relationship between Collatz trajectories and good rational approximations of $\log_2 3$ specifically, the present work generalises in both directions: the mediant tree constructs good rational approximants to *any* real θ (see [1]), and the pipeline of Section 4 produces a concrete $gx+q$ trajectory (for any $g, h \geq 2$ with $\gcd(g,h) = 1$ and $q \neq 0$) realising each approximant. The result is that every sufficiently good rational approximation to any real number is realisable by a constructible generalised Collatz trajectory.

References

[1] J. Seymour, *Rational Approximations via the Mediant Tree*, `https://github.com/wildducktheories/collatz`, Paper 76 (2026).

[2] S. Eliahou, *The* $3x + 1$ *problem: new lower bounds on nontrivial cycle lengths*, Discrete Mathematics **118** (1993), 45–56.

[3] J. Seymour, *The Structural Polynomial of a Parity Sequence*, `https://github.com/wildducktheories/collatz`, Paper 71 (2026).

[4] J. Seymour, *A Regular Expression Language for the Collatz Graph*, `https://github.com/wildducktheories/collatz`, Paper 33 (2026).

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